

Introduction: Solubility is the ability of a substance to dissolve in a solvent. It depends on the intermolecular bonds formed and changes due to temperature, pH, concentration, and polarity [1]. The importance of solubility lies within the goal of the end product as it can determine the release of bioactive substances, drug delivery kinetics, and compatibility with biological environments [2]. This experiment uses dextrose, bovine serum albumin (BSA), and poly lactic-co-glycolic acid (PLGA) as solutes: dextrose is the primary energy source of organisms, BSA is a protein derived from cow's blood, and PLGA is a biodegradable polymer that is commonly used in drug delivery. The solvents used are deionized water (DI H₂O), the commonly used solvent dimethyl sulfoxide (DMSO), and dichloromethane (DCM). The aim of this study is to find the solubility of these substances by understanding the intermolecular forces that take part in them. The final DMSO and PLGA mixture is later altered to observe any noticeable changes. It is hypothesized that DI H₂O will dissolve dextrose and BSA and DMSO will dissolve all three, both due to hydrogen bond formation and DCM will only dissolve PLGA due to Van der Waals force domination.

Methods: In the first experiment, DMSO was pipetted into three beakers which were labeled for the three solutes. A spatula was used to scoop out and weigh dextrose on an electronic balance. This was repeated with BSA and PLGA in separate weighing boats and all three solutes were added to their DMSO beakers along with magnetic stirrer pellets to create mixtures of concentration 3.33 mg/cm³. They were then placed in the magnetic stirrer for 15 minutes. After this, the solutions were observed. This was repeated for DCM and DI H₂O instead of DMSO. Due to the toxicity of DCM and DMSO, experiments with these two solvents are carried out in the fume hood. After completion, drops of DI H₂O were added to the PLGA and DMSO solution using a syringe. Upon observation of the mixture, more DI H₂O was added until no significant changes were observed.

Results: Table 1 shows that none of the solvents can dissolve all the solutes. Solubility is deduced by the complete dissolution of the substance in the solvent, forming a clear and homogenous solution. Insolubility is deduced by the lack of dissolution leading to forming precipitates in a cloudy mixture. On the other hand, after adding DI H₂O to PLGA and DMSO mixture, the mixture dissolves at first, then the temperature rises and with the addition of more DI H₂O, precipitate starts to appear, creating an immiscible solution.

Solubility chart	DI H ₂ O	DMSO	DCM
Dextrose	✓	✓	✗
BSA	✓	✗	✗
PLGA	✗	✓	✓

Discussion: Solubility lies within the intermolecular forces between solute and solvent and these forces can be dictated by hydrogen bondings, dipole-dipole interactions, or van der Waals forces [3]. H₂O dissolves both dextrose and BSA as all three are polar molecules (have uneven charge distribution due to electronegativity) resulting in hydrogen bonding, aligning with the hypothesis. Hydrogen bonding is a strong dipole-dipole interaction (when oppositely partial charges in two molecules have a force of attraction) between a hydrogen (H) atom bonded to a highly electronegative atom with another electronegative atom. This results in a partial positive charge on the H atom and a partial negative charge on the electronegative atom, resulting in a strong force of attraction. In this case, both dextrose and BSA have highly electronegative atoms (N, O) which form hydrogen bonds with H atoms in H₂O. PLGA contains many nonpolar (even charge distribution) functional groups, like carbonyl groups, and has a big molecular size that does not establish hydrogen bonds with H₂O hence they are insoluble. DMSO dissolves dextrose and PLGA through its ability to form hydrogen bonds and dipole-dipole interactions as dextrose has polar hydroxyl groups, while PLGA contains polar ester groups and polar O and H atoms which form hydrogen bonds with DMSO. BSA, although having polar O, N, and Si atoms, is a large, complex protein, less compatible with DMSO due to its complexity, so it doesn't dissolve as readily. Lastly, DCM is a nonpolar solvent that cannot dissolve dextrose and BSA due to the lack of formation of hydrogen bonding. DCM can dissolve PLGA as the nonpolar groups create Van der Waals forces, these are weak attractive forces that exist between all molecules but predominate in this case due to the lack of other intermolecular forces. The addition of DI H₂O to the PLGA and DMSO mixture caused an exothermic (heat-releasing) reaction between H₂O and DMSO, which explains the temperature rise. Adding more H₂O molecules resulted in no more DMSO molecules to react with, which led to an insoluble mixture of PLGA and H₂O as noticed previously. Ultimately, the results supported the hypothesis for DI H₂O and DCM but failed to predict the insolubility of BSA in DMSO.

Conclusion: Investigating the solubility in this experiment shows the role of intermolecular forces in solubility. Further experimentation can increase our knowledge of such attraction forces in choosing materials for biological environments.

References: [1] JX et al: Biochemistry, Dissolution, and Solubility [2] A Hubbell: Bioactive Materials [3] Burke: Solubility Parameters